Abd Elmouhib Souri

Data Science & AI Engineering Student (LLM/RAG, NLP, Computer Vision) — 4–6 month internship (PFE) from February 2027

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Summary

Final-year engineering student in Data Science and Artificial Intelligence who builds machine learning, deep learning, natural language processing (NLP), large language model (LLM) / retrieval-augmented generation (RAG) and computer vision systems end to end: data pipeline, model, API, database, Docker, automated tests. I fix the success criteria before each experiment and evaluate on data the model has never seen, so my numbers hold up; my systems refuse rather than guess. Looking for an internship in AI, data science or software engineering that can lead to a permanent role.

Technical skills

LLMs, RAG & agents: LangChain, LangGraph, RAG, AI agent pipelines, vector database (ChromaDB), BM25, cross-encoder reranking, embeddings, prompt engineering, prompt-injection defence, Groq / Gemini APIs, LLM evaluation
Machine learning & NLP: Python, scikit-learn, PyTorch, TensorFlow/Keras, Hugging Face Transformers, deep learning (CNN, Bi-LSTM), transfer learning and fine-tuning, text classification, language identification, Arabic text normalization, evaluation (held-out tests, leakage audits, confidence intervals)
Computer vision: object detection (YOLO, YOLOv8 / Ultralytics), image classification (ResNet-50, EfficientNet), OCR (TrOCR, EasyOCR / CRAFT), OpenCV
Data & BI: SQL (PostgreSQL), MongoDB, Redis, ETL, star schema, Pandas, NumPy, Power BI / DAX, statistics
Software engineering & MLOps: FastAPI, Flask, Streamlit, Gradio, Node.js / Express, React / TypeScript, Next.js, REST APIs, Docker, Git, GitHub Actions CI/CD, pytest, JWT / 2FA, Agile / Scrum

Education

Engineering degree (Bac+5), Data Science, AI & Business Intelligence — *ESPRIM, Monastir, Tunisia (final year)* 2024 – 2027
Preparatory classes, Physics–Technology — *Gafsa, Tunisia* 2022 – 2024

Professional experience

Software & AI Engineering Intern, DotJcom Ltd — *ForgeAI, built alone, public repository* Jul – Sep 2026

- Multi-agent platform that turns a startup idea into a sourced study, an argued verdict and an execution plan: 8-node **LangGraph** pipeline, 2 **ChromaDB** retrieval stores plus web search, a 9-criterion score and an 8-role committee computed by rules, not by the LLM.
- Production features: Gemini with Groq fallback, JWT, Google sign-in and 2FA, admin console with audit log, PDF/Word export; FastAPI, Next.js, PostgreSQL, Redis, Docker Compose. 25 analyses completed in testing.
- A pre-registered test (2 strong vs 3 weak ideas) showed the score did not separate them (every idea scored 7.0); I published the result and traced the cause to fixed committee values in the code.

Mobile Development Intern, Tunisie Telecom — *React Native + Firebase app with a Gemini assistant for USSD codes* Aug – Sep 2025

Projects (all numbers measured; limits stated in each repository)

MedRAG — RAG + LLM question answering over patient records — *personal, 2 versions* github.com/Abd-Elmouhib-Souri/MedRAG

- v1: hybrid RAG (ChromaDB + BM25), query rewriting, per-patient collections, clinical alerts, PDF export; FastAPI + Streamlit. It never refused: an irrelevant passage still reached the LLM.
- v2: cross-encoder relevance gate (bge-reranker-v2-m3) chosen among 3 rerankers under a rule fixed in advance (50% → 88%), page citations, a second LLM checking grounding, prompt-injection quarantine. **Held-out test run once: 10/11 correct (95% CI 62–98%), 8/8 unanswerable questions refused**; 5.4 s median; 38 tests.

DataScoop — Alzheimer MRI decision support, deep learning — *team PFA, 2 versions* github.com/Abd-Elmouhib-Souri/DataScoop

- v1: trained ResNet-50, a custom CNN and EfficientNet (ResNet-50: 0.900 accuracy); built the Node/Express backend and Flask ML service of a doctor/patient web app.
- v2: proved data leakage on 50,400 images (duplicates and augmented copies across the split) and rebuilt: **0.900** → **0.814** (95% CI 0.800–0.826) on 3,153 leak-free images; simulated predictions removed; the doctor validates every result; 45 tests.

MedBox Reader — object detection + OCR on medicine boxes — *personal, 2 versions* github.com/Abd-Elmouhib-Souri/MedBox

- v1: YOLOv8n (PyTorch, Kaggle test mAP@0.5 0.667) finds the name line, TrOCR (Hugging Face) reads it, fuzzy matching on 2,160 Tunisian trade names, refusal threshold fixed on dev. **Frozen tests, 27 images: 0 wrong answers but no drug recognized**; error analysis: training labels sit on the molecule, not the brand.
- v2: CRAFT text detection on the tallest lines, brand or molecule answer or refusal: 6/8 brands, 0/13 wrong on dev photos (thresholds tuned there, not a test); independent test pre-registered. Gradio demo, 16 tests, CI.

MindCare AI — distress-language detection, NLP — *personal, 2 versions* github.com/Abd-Elmouhib-Souri/MindCare-AI

- v1: multimodal prototype (face FER-2013, voice MFCC, text Bi-LSTM + GloVe) reporting 88% text recall. v2: my audit found that 88% came from 50 sentences copied 3,000 times before the split; rebuilt with a hash split, test run once: **Bi-LSTM macro-F1 0.938 vs 0.936 for TF-IDF** on 34,635 posts; unseen-speaker voice accuracy 0.453 (was 0.981).

NewsCheck TN and 9anoun AI — Arabic/Tunisian NLP — *team projects in progress, 9anoun with the startup Tradrly* private repos

- NewsCheck, my part: language and script identification (French, MSA, Tunisian in Arabic script or Arabizi): **macro-F1 0.94 vs 0.50 baseline** on a 34-sentence frozen test; 0/680 MSA sentences mislabelled; recall 0.53 on real comments, so a MARBERTv2 model is planned.
- 9anoun, my part: legal retrieval with a versioned article corpus, BM25 with Arabic normalization and an abstention rule: Recall@3 16/17, 6/8 out-of-corpus questions refused; 185 tests on my module.

Sales forecasting pipeline and Power BI reports — *personal and course* github.com/Abd-Elmouhib-Souri/sales-forecasting-pipeline

- ETL of 9,800 transactions into a PostgreSQL star schema (Docker); on held-out 2018, Ridge **MAPE 20.3% vs 24.9%** (seasonal naive). Found and fixed a wrong profit formula in a Power BI report.

Certifications & languages

Google BI Professional Certificate (Coursera, 2026) | **DataCamp** Supervised Learning with scikit-learn (2026)
Languages: Arabic (native), French (C1), English (B1, technical reading and writing)